

System of Linear Equations

A collection of one or more linear equations with same number of unknowns $x_1, \dots, x_n$

e.g. $2x_1 - x_2 + 1.5x_3 = 8$ ①
 $x_1 - 4x_3 = -7$ ②

solutions: $(x_1 = -7, x_2 = -22, x_3 = 0)$, $(x_1 = 1, x_2 = -3, x_3 = 2)$,
solution set:

$$\begin{aligned} \text{①} - 2 \times \text{②} &\Rightarrow -x_2 + 1.5x_3 - (-8x_3) = 8 - (-14) \\ &\Rightarrow -x_2 + 9.5x_3 = 22 \\ &\Rightarrow x_2 = 9.5x_3 - 22 \end{aligned}$$

$$\text{②} \Rightarrow x_1 = 4x_3 - 7$$

Then the solution set is the set of (x_1, x_2, x_3) such that
 $x_1 = 4x_3 - 7$ and $x_2 = 9.5x_3 - 22 \Rightarrow$ infinitely many solutions

A system of equations can have

{	1. no solution	inconsistent
	2. exactly one solution	consistent
	3. infinitely many solution	

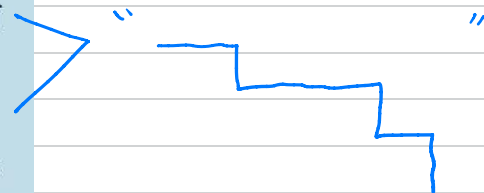
Row Reduction & Echelon Forms

A rectangular matrix is in **echelon form** (or **row echelon form**) if it has the following three properties:

1. All nonzero rows are above any rows of all zeros.
2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
3. All entries in a column below a leading entry are zeros.

If a matrix in echelon form satisfies the following additional conditions, then it is in **reduced echelon form** (or **reduced row echelon form**):

4. The leading entry in each nonzero row is 1.
5. Each leading 1 is the only nonzero entry in its column.



eg.

$$\begin{bmatrix} \textcircled{2} & -3 & 2 & 1 \\ 0 & \textcircled{1} & -4 & 8 \\ 0 & 0 & 0 & \textcircled{5/2} \end{bmatrix}$$

EF

and

$$\begin{bmatrix} \textcircled{1} & 0 & 0 & 29 \\ 0 & \textcircled{1} & 0 & 16 \\ 0 & 0 & \textcircled{1} & 3 \end{bmatrix}$$

EF & RREF

Vector Equations

Linear Combination

Given vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_p \in \mathbb{R}^n$ and given scalars $c_1, c_2, \dots, c_p \in \mathbb{R}$, the vector $\vec{y}$

$$\vec{y} = c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_p \vec{v}_p$$

is called a linear combination of $v_1, \dots, v_p$ with weights $c_1, \dots, c_p$

Span of vectors

If $v_1, \dots, v_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $v_1, \dots, v_p$ is denoted by $\text{Span}\{v_1, \dots, v_p\}$ and is called the **subset of $\mathbb{R}^n$ spanned (or generated)** by $v_1, \dots, v_p$. That is, $\text{Span}\{v_1, \dots, v_p\}$ is the collection of all vectors that can be written in the form

$$c_1 v_1 + c_2 v_2 + \dots + c_p v_p$$

with $c_1, \dots, c_p$ scalars.

The Matrix Equation $A\vec{x} = \vec{b}$

If A is an $m \times n$ matrix, with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$, and if $\mathbf{x}$ is in $\mathbb{R}^n$, then the product of A and $\mathbf{x}$, denoted by $A\mathbf{x}$, is the linear combination of the columns of A using the corresponding entries in $\mathbf{x}$ as weights; that is,

$$A\mathbf{x} = \underbrace{[\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]}_{\text{vectors}} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \underbrace{x_1}_{\text{weights}} \mathbf{a}_1 + \underbrace{x_2}_{\text{weights}} \mathbf{a}_2 + \cdots + \underbrace{x_n}_{\text{weights}} \mathbf{a}_n$$

$A\vec{x} = \vec{b}$ is consistent $\Leftrightarrow \vec{b}$ is a linear combination of columns of $A \Leftrightarrow \vec{b} \in \text{span}\{\vec{a}_1, \dots, \vec{a}_n\}$.

Solution Set of Linear System

Homogeneous system

$$\left[\begin{array}{cccc|c} * & * & - & * & 0 \\ * & * & . & * & 0 \\ * & * & - & * & 0 \end{array} \right]$$

trivial solution $\longleftrightarrow$ pivot in every column $\longleftrightarrow$ no free variable

non-trivial solution $\longleftrightarrow$ some column is not a pivot $\longleftrightarrow$ free variables

Use free variable $\Rightarrow$ parametric vector form.

Linear Independence

An indexed set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is said to be **linearly independent** if the vector equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_p\mathbf{v}_p = \mathbf{0}$$

has only the trivial solution. The set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is said to be **linearly dependent** if there exist weights $c_1, \dots, c_p$, not all zero, such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p = \mathbf{0} \quad (2)$$

e.g. $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

e.g. $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

The columns of a matrix A are linearly independent if and only if the equation $A\mathbf{x} = \mathbf{0}$ has *only* the trivial solution. (3)

$$1 \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \vec{0} \Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Linear Transformation

$$T: \mathbb{R}^n \longrightarrow \mathbb{R}^m$$

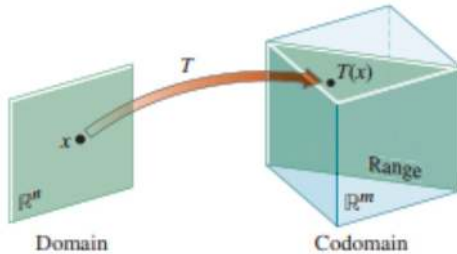


FIGURE 2 Domain, codomain, and range of $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$.

$$\text{e.g. } T(\vec{x}) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \vec{x}$$

domain: $\mathbb{R}^2$

codomain: $\mathbb{R}^2$

range: $\left\{ \begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2 \mid a=0 \right\}$

Matrix transformation

$$T(\vec{x}) = A\vec{x} \quad \text{i.e. } T: \vec{x}_1 \longrightarrow A\vec{x}$$

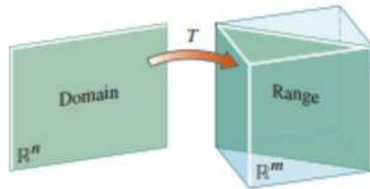
Linear transformation

A transformation (or mapping) T is **linear** if

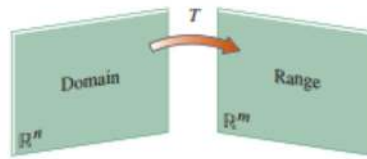
- (i) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in the domain of T ;
- (ii) $T(c\mathbf{u}) = cT(\mathbf{u})$ for all scalars c and all $\mathbf{u}$ in the domain of T .

One-to-One & Onto

A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **onto** $\mathbb{R}^m$ if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at least one* $\mathbf{x}$ in $\mathbb{R}^n$.

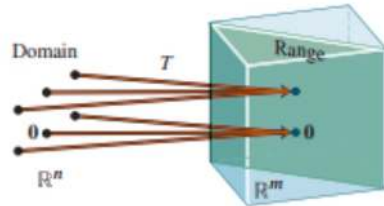


T is not onto $\mathbb{R}^m$

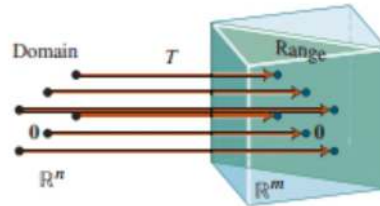


T is onto $\mathbb{R}^m$

A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **one-to-one** if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at most one* $\mathbf{x}$ in $\mathbb{R}^n$.



T is not one-to-one



T is one-to-one

$A\vec{x} = \vec{b}$ has a solution for any $\vec{b}$

$\Rightarrow$ If row reduce A , we don't

$$\text{want } \begin{bmatrix} * & * & * \\ * & * & * \\ 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow$ pivot in every row

$A\vec{x} = \vec{b}$ has unique solution in particular

$A\vec{x} = \vec{0}$ has only trivial solution

$\Rightarrow$ pivot in every column.

THEOREM 10

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that

$$T(\mathbf{x}) = A\mathbf{x} \quad \text{for all } \mathbf{x} \text{ in } \mathbb{R}^n$$

In fact, A is the $m \times n$ matrix whose j th column is the vector $T(\mathbf{e}_j)$, where $\mathbf{e}_j$ is the j th column of the identity matrix in $\mathbb{R}^n$:

$$A = [T(\mathbf{e}_1) \quad \cdots \quad T(\mathbf{e}_n)] \quad (3)$$

$$\text{If } A = [\vec{a}_1 \quad \vec{a}_2 \quad \vec{a}_3 \quad \cdots \quad \vec{a}_n]$$

$$T(\vec{e}_1) = \vec{a}_1, \quad \cdots, \quad T(\vec{e}_n) = \vec{a}_n$$

Matrix Operation

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$$

$$AB = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} \end{bmatrix}$$

$$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \quad C^2 = C \cdot C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$$

C & D commute if $CD = DC$
↑ ↑
square matrices

Inverse of a Matrix

An $n \times n$ matrix A is said to be **invertible** if there is an $n \times n$ matrix C such that

$$\begin{array}{c} \overline{CA} = I \quad \text{and} \quad \overline{AC} = I \\ \uparrow \qquad \qquad \uparrow \\ A^{-1} \qquad \quad A^{-1} \end{array}$$

THEOREM 4

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If $ad - bc = 0$, then A is not invertible.

Elementary Matrices

An **elementary matrix** is one that is obtained by performing a single elementary row operation on an identity matrix. The next example illustrates the three kinds of elementary matrices.

EXAMPLE 5

Let

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix},$$
$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Compute E_1A , E_2A , and E_3A , and describe how these products can be obtained by elementary row operations on A .

SOLUTION Verify that

$$E_1A = \begin{bmatrix} a & b & c \\ d & e & f \\ g - 4d & h - 4e & i - 4f \end{bmatrix}, \quad E_2A = \begin{bmatrix} d & e & f \\ a & b & c \\ g & h & i \end{bmatrix},$$
$$E_3A = \begin{bmatrix} a & b & c \\ d & e & f \\ 5g & 5h & 5i \end{bmatrix}.$$

Find A^{-1}

$$[A | I] \xrightarrow[\substack{E_1 E_2 \dots E_n}]{RR} [I | A^{-1}]$$

Characterizations of Invertible Matrices

THEOREM 8

The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix. Then the following statements are equivalent. That is, for a given A , the statements are either all true or all false.

- A is an invertible matrix.
- A is row equivalent to the $n \times n$ identity matrix.
- A has n pivot positions.
- The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- The columns of A form a linearly independent set.
- The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
- The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$.
- The columns of A span $\mathbb{R}^n$.
- The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
- There is an $n \times n$ matrix C such that $CA = I$.
- There is an $n \times n$ matrix D such that $AD = I$.
- A^T is an invertible matrix.

Partitioned Matrices

EXAMPLE 1 The matrix

$$A = \left[\begin{array}{ccc|cc|c} 3 & 0 & -1 & 5 & 9 & -2 \\ -5 & 2 & 4 & 0 & -3 & 1 \\ -8 & -6 & 3 & 1 & 7 & -4 \end{array} \right]$$

can also be written as the 2×3 **partitioned** (or **block**) matrix

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix}$$

whose entries are the *blocks* (or *submatrices*)

$$A_{11} = \begin{bmatrix} 3 & 0 & -1 \\ -5 & 2 & 4 \end{bmatrix}, \quad A_{12} = \begin{bmatrix} 5 & 9 \\ 0 & -3 \end{bmatrix}, \quad A_{13} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$A_{21} = \begin{bmatrix} -8 & -6 & 3 \end{bmatrix}, \quad A_{22} = \begin{bmatrix} 1 & 7 \end{bmatrix}, \quad A_{23} = \begin{bmatrix} -4 \end{bmatrix}$$

• Addition & scalar multiplication are as ordinary sum and scalar multiplication

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} + \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11}+B_{11} & A_{12}+B_{12} \\ A_{21}+B_{21} & A_{22}+B_{22} \end{bmatrix}$$

$$c \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} cA_{11} & cA_{12} \\ cA_{21} & cA_{22} \end{bmatrix}$$

Multiplication of Partitioned Matrices

Partitioned matrices can be multiplied by the usual row-column rule as if the block entries were scalars, provided that for a product AB , the column partition of A matches the row partition of B .

EXAMPLE 3 Let

$$A = \left[\begin{array}{ccc|cc} 2 & -3 & 1 & 0 & -4 \\ 1 & 5 & -2 & 3 & -1 \\ 0 & -4 & -2 & 7 & -1 \end{array} \right] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \left[\begin{array}{cc} 6 & 4 \\ -2 & 1 \\ -3 & 7 \\ -1 & 3 \\ 5 & 2 \end{array} \right] = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

The 5 columns of A are partitioned into a set of 3 columns and then a set of 2 columns. The 5 rows of B are partitioned in the same way—into a set of 3 rows and then a set of 2 rows. We say that the partitions of A and B are **conformable** for **block multiplication**. It can be shown that the ordinary product AB can be written as

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = \begin{bmatrix} A_{11}B_1 + A_{12}B_2 \\ A_{21}B_1 + A_{22}B_2 \end{bmatrix} = \begin{bmatrix} -5 & 4 \\ -6 & 2 \\ 2 & 1 \end{bmatrix}$$

It is important for each smaller product in the expression for AB to be written with the submatrix from A on the left, since matrix multiplication is not commutative. For instance,

$$\begin{aligned} A_{11}B_1 &= \begin{bmatrix} 2 & -3 & 1 \\ 1 & 5 & -2 \end{bmatrix} \begin{bmatrix} 6 & 4 \\ -2 & 1 \\ -3 & 7 \end{bmatrix} = \begin{bmatrix} 15 & 12 \\ 2 & -5 \end{bmatrix} \\ A_{12}B_2 &= \begin{bmatrix} 0 & -4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} -20 & -8 \\ -8 & 7 \end{bmatrix} \end{aligned}$$

Hence the top block in AB is

$$A_{11}B_1 + A_{12}B_2 = \begin{bmatrix} 15 & 12 \\ 2 & -5 \end{bmatrix} + \begin{bmatrix} -20 & -8 \\ -8 & 7 \end{bmatrix} = \begin{bmatrix} -5 & 4 \\ -6 & 2 \end{bmatrix}$$



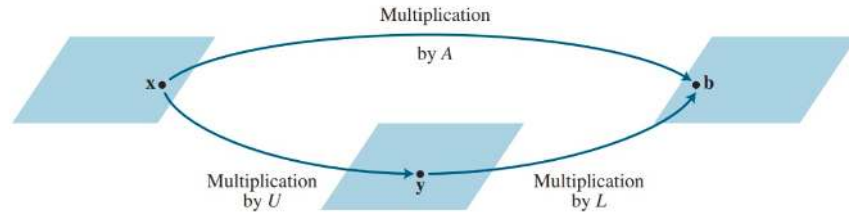
Matrix Factorization

LU Factorization:

Let matrix A be an $m \times n$ matrix, the LU factorization gives a way to rewrite $A = LU$ where L is an $m \times m$ lower triangular matrix and U is an $m \times n$ echelon form of A

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix} \begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

L U



Using LU factorization to solve: ① $L\vec{y} = \vec{b}$: $[L \vec{b}] \longrightarrow [I \vec{y}]$
② $U\vec{x} = \vec{y}$

Subspaces of $\mathbb{R}^n$

A **subspace** of $\mathbb{R}^n$ is any set H in $\mathbb{R}^n$ that has three properties:

- The zero vector is in H .
- For each $\mathbf{u}$ and $\mathbf{v}$ in H , the sum $\mathbf{u} + \mathbf{v}$ is in H .
- For each $\mathbf{u}$ in H and each scalar c , the vector $c\mathbf{u}$ is in H .

$A: m \times n$

Examples:

- **Column space** of a matrix A ($\text{Col } A$) contains all linear combinations of columns of A (subspace of $\mathbb{R}^m$)
- **Null space** of a matrix A ($\text{Nul } A$) contains all solutions of the homogeneous equation $A\vec{x} = \vec{0}$ (subspace of $\mathbb{R}^n$)

A **basis** for a subspace H of $\mathbb{R}^n$ is a linearly independent set in H that spans H .

basis of $\text{Col } A$: pivot columns

basis of $\text{Nul } A$: from the parametric vector form.

dimension: number of vectors in a basis

Dimension and Rank

The **dimension** of a nonzero subspace H , denoted by $\dim H$, is the number of vectors in any basis for H . The dimension of the zero subspace $\{0\}$ is defined to be zero.²

The **rank** of a matrix A , denoted by $\text{rank } A$, is the dimension of the column space of A .

The Rank Theorem

If a matrix A has n columns, then $\text{rank } A + \dim \text{Nul } A = n$.

The Invertible Matrix Theorem (continued)

Let A be an $n \times n$ matrix. Then the following statements are each equivalent to the statement that A is an invertible matrix.

- m. The columns of A form a basis of $\mathbb{R}^n$. $\Rightarrow$ pivot in every column
- n. $\text{Col } A = \mathbb{R}^n \Rightarrow m$.
- o. $\text{rank } A = n \Rightarrow m$.
- p. $\dim \text{Nul } A = 0 \Rightarrow$ no free variable $\Rightarrow$ pivot in every column
- q. $\text{Nul } A = \{0\} \Rightarrow p$.

Alternating Sign Pattern:

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \\ + & - & + & \\ \vdots & & & \ddots \end{bmatrix}$$

Introduction to Determinants

For $n \geq 2$, the **determinant** of an $n \times n$ matrix $A = [a_{ij}]$ is the sum of n terms of the form $\pm a_{1j} \det A_{1j}$, with plus and minus signs alternating, where the entries $a_{11}, a_{12}, \dots, a_{1n}$ are from the first row of A . In symbols,

$$\begin{aligned} \det A &= \underline{a_{11} \det A_{11}} - \underline{a_{12} \det A_{12}} + \cdots + \underline{(-1)^{1+n} a_{1n} \det A_{1n}} \\ &= \sum_{j=1}^n (-1)^{1+j} a_{1j} \det A_{1j} \end{aligned}$$

← cofactor expansion across the first row.

$$(i,j)\text{-cofactor} = C_{ij} = (-1)^{i+j} a_{ij} \det(A_{ij})$$

Theorems:

If A is a triangular matrix, then $\det A$ is the product of the entries on the main diagonal of A .

A square matrix A is invertible if and only if $\det A \neq 0$. (A is nonsingular)

Example of finding A_{ij}

$$A = \begin{bmatrix} 1 & -2 & 5 & 0 \\ 2 & 0 & 4 & -1 \\ 3 & 1 & 0 & 7 \\ 0 & 4 & -2 & 0 \end{bmatrix}$$

then A_{32} is obtained by crossing out row 3 and column 2,

$$\begin{bmatrix} 1 & -2 & 5 & 0 \\ 2 & 0 & 4 & -1 \\ 3 & 1 & 0 & 7 \\ 0 & 4 & -2 & 0 \end{bmatrix}$$

so that

$$A_{32} = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

Properties of Determinants

$$(b) B = \begin{pmatrix} 0 & 1 & t \\ -3 & 10 & 0 \\ 0 & 5 & k \end{pmatrix} \xrightarrow{?} \begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{bmatrix}$$

Row Operations

Let A be a square matrix.

- If a multiple of one row of A is added to another row to produce a matrix B , then $\det B = \det A$.
- If two rows of A are interchanged to produce B , then $\det B = -\det A$.
- If one row of A is multiplied by k to produce B , then $\det B = k \det A$.

If A is an $n \times n$ matrix, then $\det A^T = \det A$. $\Rightarrow$ row operations

Multiplicative Property

If A and B are $n \times n$ matrices, then $\det AB = (\det A)(\det B)$.

Now we have two general techniques for computing determinants of $n \times n$ matrix A ($n \geq 2$)

Example:

$$A = \begin{bmatrix} 0 & 1 & 2 \\ -3 & 10 & 0 \\ 0 & 5 & 1 \end{bmatrix}$$

[Method 1]: (Cofactor Expansion)

$$\det(A) = (-1)^{2+1} (-3) \det \begin{bmatrix} 1 & 2 \\ 5 & 1 \end{bmatrix} = 3 \cdot (1 - 10) = -27$$

[Method 2]: (Row Operations)

$$A = \begin{bmatrix} 0 & 1 & 2 \\ -3 & 10 & 0 \\ 0 & 5 & 1 \end{bmatrix} \xrightarrow[\text{change sign}]{R_1 \leftrightarrow R_2} \begin{bmatrix} -3 & 10 & 0 \\ 0 & 1 & 2 \\ 0 & 5 & 1 \end{bmatrix} \xrightarrow[\text{no change}]{R_3 \rightarrow R_3 - 5R_2} \begin{bmatrix} -3 & 10 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & -9 \end{bmatrix}$$

$$\det(A) = -\det \begin{bmatrix} -3 & 10 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & -9 \end{bmatrix} = -1((-3) \times 1 \times (-9)) = -27$$

Cramer's Rule, Volume, and Linear Transformation

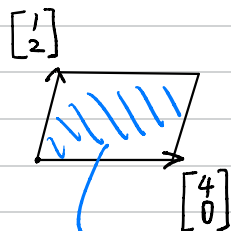
Cramer's Rule

Let A be an invertible $n \times n$ matrix. For any $\mathbf{b}$ in $\mathbb{R}^n$, the unique solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ has entries given by

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}, \quad i = 1, 2, \dots, n \quad (1)$$

Determinant as Area or Volume

If A is a 2×2 matrix, the area of the parallelogram determined by the columns of A is $|\det A|$. If A is a 3×3 matrix, the volume of the parallelepiped determined by the columns of A is $|\det A|$.



$$\text{Area} = \left| \det \begin{bmatrix} 1 & 4 \\ 2 & 0 \end{bmatrix} \right|$$

Linear Transformations

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation determined by a 2×2 matrix A . If S is a parallelogram in $\mathbb{R}^2$, then

$$\{\text{area of } T(S)\} = |\det A| \cdot \{\text{area of } S\} \quad (5)$$

If T is determined by a 3×3 matrix A , and if S is a parallelepiped in $\mathbb{R}^3$, then

$$\{\text{volume of } T(S)\} = |\det A| \cdot \{\text{volume of } S\} \quad (6)$$

Application to Markov Chains

A vector with nonnegative entries that add up to 1 is called a **probability vector**.

A **stochastic matrix** is a square matrix whose columns are probability vectors.

A **Markov chain** is a sequence of probability vectors $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \dots$, together with a stochastic matrix P , such that

$$\mathbf{x}_1 = P\mathbf{x}_0, \quad \mathbf{x}_2 = P\mathbf{x}_1, \quad \mathbf{x}_3 = P\mathbf{x}_2, \quad \dots$$

Thus the Markov chain is described by the first-order difference equation

$$\mathbf{x}_{k+1} = P\mathbf{x}_k \quad \text{for } k = 0, 1, 2, \dots$$

$\mathbf{x}_i$ is called a state vector $i=1, 2, \dots$

Regular stochastic matrix is when P^2 has every entry being positive.

Eigenvectors & Eigenvalues

An **eigenvector** of an $n \times n$ matrix A is a nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ . A scalar λ is called an **eigenvalue** of A if there is a nontrivial solution $\mathbf{x}$ of $A\mathbf{x} = \lambda\mathbf{x}$; such an $\mathbf{x}$ is called an *eigenvector corresponding to λ* .¹

↑ Eigenvector
↗ Eigenvalue

$$A\vec{x} = \lambda\vec{x} \Rightarrow A\vec{x} - \lambda\vec{x} = 0 \Rightarrow (A - \lambda I)\vec{x} = 0$$

If there exists a nonzero eigenvector $\vec{x}$ corresponding to the eigenvalue λ then $\det(A - \lambda I) = 0$

The Characteristic Equation

A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if λ satisfies the characteristic equation

$$\det(A - \lambda I) = 0$$

The general idea of find eigenvalues and eigenvectors:

- ① define the characteristic eqn: $\det(A - \lambda I) = 0$
- ② Solve for the eigenvalues (solve the characteristic eqn).
 $\lambda_1, \dots, \lambda_n$
- ③ for each $\lambda_1, \dots, \lambda_n$ find the basis of the null space of $A - \lambda_i I$
 $[A - \lambda_1 I | 0], [A - \lambda_2 I | 0], \dots, [A - \lambda_n I]$

Diagonalization

Q: Let $D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$, $D^3 = ?$ $\begin{bmatrix} 5^3 & 0 \\ 0 & 3^3 \end{bmatrix}$

Q: Let $A = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$, $A = PDP^{-1} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}^{-1}$, $A^3 = ?$

$$\begin{aligned} A^3 &= (PDP^{-1})(PDP^{-1})(PDP^{-1}) \\ &= PDP^{-1}PDP^{-1}PDP^{-1} \\ &= PD D P P^{-1} \\ &= PD^3 P^{-1} \\ &= \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1^3 & 0 \\ 0 & 2^3 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}^{-1} \end{aligned}$$

The Diagonalization Theorem

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

In fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A . In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P .

$$\underset{\substack{\uparrow \\ n \times n}}{A} = \underbrace{\begin{bmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{bmatrix}}_P \underbrace{\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}}_D \underbrace{\begin{bmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{bmatrix}^{-1}}_{P^{-1}}$$

If A is diagonalizable, to find diagonalization:

- ① Find all eigenvalues & corresponding eigenvectors
- ② Place eigenvalues on the diagonals of D and eigenvectors in cols of P

Diagonalizability:

A diagonalizable $\Leftrightarrow$ A has n linearly independent eigenvectors
(When algebraic multiplicities = geometric multiplicities)

Always true: geometric multiplicities $\leq$ algebraic multiplicities = n

Example:

$$\text{Let } A = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 5 \end{bmatrix}$$

eigenvalues: $\underline{3, 3, 5}$ ← algebraic multiplicity = 1.

↑
algebraic multiplicity = 2

$$\Rightarrow (3-\lambda)^2 (5-\lambda)^1 = 0$$

characteristic equation: $\det(A - \lambda I) = 0 \Rightarrow \det \begin{bmatrix} 3-\lambda & 1 & 2 \\ 0 & 3-\lambda & 3 \\ 0 & 0 & 5-\lambda \end{bmatrix} = 0$

λ -eigenspace = $\text{Nul}(A - \lambda I)$

free variables in $A - \lambda I = \dim \text{Nul}(A - \lambda I)$

$$A - 3I = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$\dim \text{Nul}(A - 3I) = \dim 3\text{-eigenspace} = 1$
= # L.I. eigenvectors

← geometric multiplicity = 1

↑ free
↑ pivots

$$A - 5I = \begin{bmatrix} -2 & 1 & 2 \\ 0 & -2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$\dim \text{Nul}(A - 5I) = \dim 5\text{-eigenspace} = 1$
= # L.I. eigenvectors

← geometric multiplicity = 2

↑ pivots
↑ free

Geometric Multiplicities < Algebraic Multiplicities
 $2+1 < 1+1$

Practice Problem

Show that if a and b are real, then the eigenvalues of $A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ are $a \pm bi$, with corresponding eigenvectors $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ i \end{bmatrix}$.

Complex Eigenvalues

Ex: Let $A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \Rightarrow \det(A - \lambda I) = \lambda^2 - \text{tr}(A)\lambda + \det(A)$
 $= \lambda^2 - 2a\lambda + (a^2 + b^2) = 0$

$$\lambda = \frac{2a \pm \sqrt{4a^2 - 4(a^2 + b^2)}}{2}$$

$$\lambda = \frac{2a \pm \sqrt{4a^2 - 4a^2 - 4b^2}}{2}$$

$$\lambda = \frac{2a \pm 2\sqrt{-b^2}}{2} = a \pm bi$$

eigenvectors: $A - (a+bi)I = \begin{bmatrix} -bi & -b \\ b & -bi \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -i \\ 0 & 0 \end{bmatrix}$

$$x_1 = ix_2 \Rightarrow \vec{x} = x_2 \begin{bmatrix} i \\ 1 \end{bmatrix} = c \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

$$A - (a-bi)I = \begin{bmatrix} bi & -b \\ b & bi \end{bmatrix} \rightarrow \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix}$$

$$\vec{x} = x_2 \begin{bmatrix} -i \\ 1 \end{bmatrix} = c \begin{bmatrix} 1 \\ i \end{bmatrix}$$

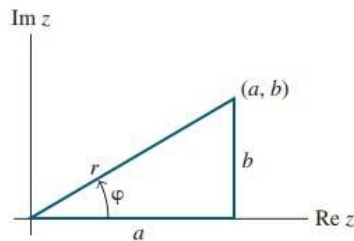


FIGURE 2

EXAMPLE 6 If $C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, where a and b are real and not both zero, then the eigenvalues of C are $\lambda = a \pm bi$. (See the Practice Problem at the end of this section.) Also, if $r = |\lambda| = \sqrt{a^2 + b^2}$, then

$$C = r \begin{bmatrix} a/r & -b/r \\ b/r & a/r \end{bmatrix} = \begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix}$$

← scaling ← rotation

where φ is the angle between the positive x -axis and the ray from $(0, 0)$ through (a, b) . See Figure 2 and Appendix B. The angle φ is called the *argument* of $\lambda = a + bi$. Thus

$$\text{Let } \lambda = a + bi, \quad \bar{\lambda} = a - bi$$

↖ conjugate

$$\text{Let } Ax = \lambda x$$

$$\text{then } A\bar{x} = \overline{A\bar{x}} = \overline{Ax} = \overline{\lambda x} = \bar{\lambda} \bar{x}$$

$$\text{Hence } A\bar{x} = \bar{\lambda} \bar{x}$$

Thus if λ is an eigenvalue of A , then $\bar{\lambda}$ is also an eigenvalue of A

Inner Product, Length, and Orthogonality

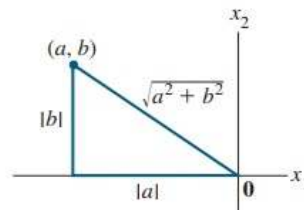
Inner Product ($\vec{u} \cdot \vec{v}$)

$$\vec{u} \cdot \vec{v} = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

Length ($\|\vec{u}\|$)

The **length** (or **norm**) of $\mathbf{v}$ is the nonnegative scalar $\|\mathbf{v}\|$ defined by

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}, \quad \text{and} \quad \|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$$



Orthogonality

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are **orthogonal** (to each other) if $\mathbf{u} \cdot \mathbf{v} = 0$.

Orthogonal Component ($W^\perp$)

To provide practice using inner products, we introduce a concept here that will be of use in Section 6.3 and elsewhere in the chapter. If a vector $\mathbf{z}$ is orthogonal to every vector in a subspace W of $\mathbb{R}^n$, then $\mathbf{z}$ is said to be **orthogonal to W** . The set of all vectors $\mathbf{z}$ that are orthogonal to W is called the **orthogonal complement** of W and is denoted by $W^\perp$ (and read as “ W perpendicular” or simply “ W perp”).

$$W^\perp = \{ \mathbf{z} \in \mathbb{R}^n \mid \mathbf{z} \cdot \mathbf{w} = 0 \quad \forall \mathbf{w} \in W \}$$

Let A be an $m \times n$ matrix. The orthogonal complement of the row space of A is the null space of A , and the orthogonal complement of the column space of A is the null space of A^T :

$$(\text{Row } A)^\perp = \text{Nul } A \quad \text{and} \quad (\text{Col } A)^\perp = \text{Nul } A^T$$

Orthogonal Sets

Let $S = \{u_1, u_2, \dots, u_p\}$ is a set of orthogonal vectors, called an orthogonal set.

An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.

Let $\{u_1, \dots, u_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each y in W , the weights in the linear combination

$$y = c_1 u_1 + \dots + c_p u_p$$

are given by

$$c_j = \frac{y \cdot u_j}{u_j \cdot u_j} \quad (j = 1, \dots, p)$$

Orthogonal Projection

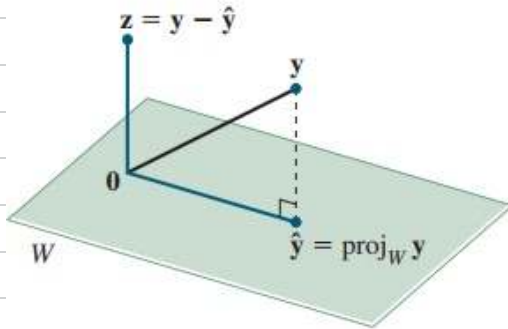


FIGURE 2

Finding α to make $y - \hat{y}$ orthogonal to u .

$$\hat{y} = \text{proj}_L y = \frac{y \cdot u}{u \cdot u} u$$

Orthonormal Set

- Orthogonal set
- All vectors in the set have unit length (length=1)

An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I$.

Orthogonal Projections

Recall from Last time:

Orthogonal Projection

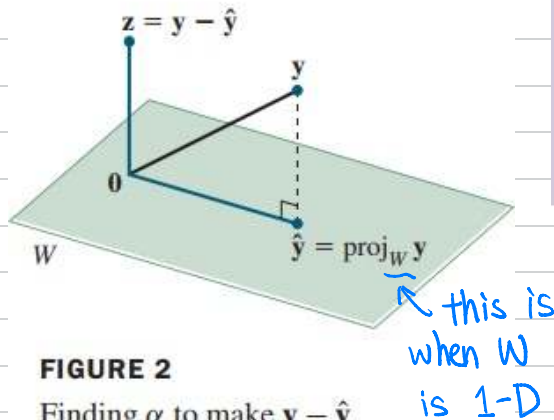


FIGURE 2

Finding α to make $y - \hat{y}$ orthogonal to u .

$$\hat{y} = \text{proj}_L y = \frac{y \cdot u}{u \cdot u} u$$

The Orthogonal Decomposition Theorem

Let W be a subspace of $\mathbb{R}^n$. Then each y in $\mathbb{R}^n$ can be written uniquely in the form

$$y = \hat{y} + z \quad (1)$$

where $\hat{y}$ is in W and z is in $W^\perp$. In fact, if $\{u_1, \dots, u_p\}$ is any orthogonal basis of W , then

$$\hat{y} = \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \dots + \frac{y \cdot u_p}{u_p \cdot u_p} u_p \quad (2)$$

and $z = y - \hat{y}$.

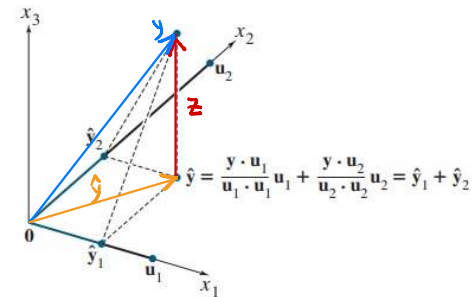


FIGURE 3 The orthogonal projection of y is the sum of its projections onto one-dimensional subspaces that are mutually orthogonal.

Gram-Schmidt Process

The Gram-Schmidt Process

Given a basis $\{\mathbf{x}_1, \dots, \mathbf{x}_p\}$ for a nonzero subspace W of $\mathbb{R}^n$, define

$$\mathbf{v}_1 = \mathbf{x}_1$$

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1$$

$$\mathbf{v}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2$$

$$\vdots$$

$$\mathbf{v}_p = \mathbf{x}_p - \frac{\mathbf{x}_p \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 - \frac{\mathbf{x}_p \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 - \dots - \frac{\mathbf{x}_p \cdot \mathbf{v}_{p-1}}{\mathbf{v}_{p-1} \cdot \mathbf{v}_{p-1}} \mathbf{v}_{p-1}$$

Then $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is an orthogonal basis for W . In addition

$$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \text{Span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\} \quad \text{for } 1 \leq k \leq p \quad (1)$$

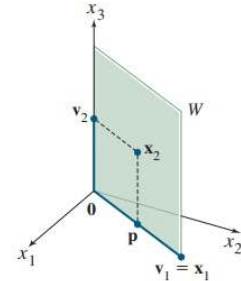


FIGURE 1

Construction of an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$.

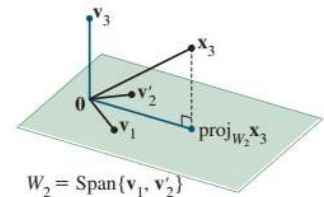


FIGURE 2 The construction of $\mathbf{v}_3$ from $\mathbf{x}_3$ and W_2 .

QR Factorization

The QR Factorization

If A is an $m \times n$ matrix with linearly independent columns, then A can be factored as $A = QR$, where Q is an $m \times n$ matrix whose columns form an orthonormal basis for $\text{Col } A$ and R is an $n \times n$ upper triangular invertible matrix with positive entries on its diagonal.

- columns of Q can be found by the Gram-Schmidt Process, then normalize
- Since columns of Q are orthogonal, then $Q^T = Q^{-1}$, so that $R = Q^T A$.

Least-Squares Problems

If A is $m \times n$ and $\mathbf{b}$ is in $\mathbb{R}^m$, a **least-squares solution** of $A\mathbf{x} = \mathbf{b}$ is an $\hat{\mathbf{x}}$ in $\mathbb{R}^n$ such that

$$\|\mathbf{b} - A\hat{\mathbf{x}}\| \leq \|\mathbf{b} - A\mathbf{x}\|$$

for all $\mathbf{x}$ in $\mathbb{R}^n$.

The least-squares solution $\hat{\mathbf{x}}$ is obtained when $\hat{\mathbf{b}} = A\hat{\mathbf{x}}$ is the projection of $\mathbf{b}$ onto $\text{Col } A$.

The set of least-squares solutions of $A\mathbf{x} = \mathbf{b}$ coincides with the nonempty set of solutions of the **normal equations** $A^T A \mathbf{x} = A^T \mathbf{b}$.

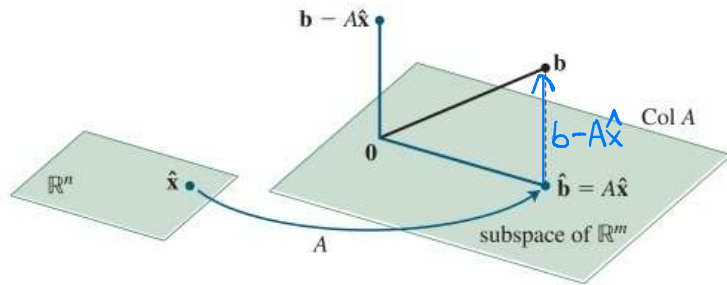


FIGURE 2 The least-squares solution $\hat{\mathbf{x}}$ is in $\mathbb{R}^n$.

$$A^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0 \Rightarrow A^T \mathbf{b} - A^T A \hat{\mathbf{x}} = 0 \Rightarrow A^T A \hat{\mathbf{x}} = A^T \mathbf{b} \quad A\hat{\mathbf{x}} = QR\hat{\mathbf{x}} = QRR^{-1}Q^T \mathbf{b} \\ = QQ^T \mathbf{b} = \text{Proj}_{\text{Col } Q} \mathbf{b} = \text{Proj}_{\text{Col } A} \mathbf{b} = \hat{\mathbf{b}}$$

Let A be an $m \times n$ matrix. The following statements are logically equivalent:

- The equation $A\mathbf{x} = \mathbf{b}$ has a unique least-squares solution for each $\mathbf{b}$ in $\mathbb{R}^m$.
- The columns of A are linearly independent.
- The matrix $A^T A$ is invertible.

When these statements are true, the least-squares solution $\hat{\mathbf{x}}$ is given by

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b} \quad (4)$$

Given an $m \times n$ matrix A with linearly independent columns, let $A = QR$ be a QR factorization of A as in Theorem 12. Then, for each $\mathbf{b}$ in $\mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a unique least-squares solution, given by

$$\hat{\mathbf{x}} = R^{-1} Q^T \mathbf{b} \quad (6)$$

Page Rank

Recall:

A vector with nonnegative entries that add up to 1 is called a **probability vector**. A **stochastic matrix** is a square matrix whose columns are probability vectors.

A **Markov chain** is a sequence of probability vectors $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \dots$, together with a stochastic matrix P , such that

$$\mathbf{x}_1 = P\mathbf{x}_0, \quad \mathbf{x}_2 = P\mathbf{x}_1, \quad \mathbf{x}_3 = P\mathbf{x}_2, \quad \dots$$

Thus the Markov chain is described by the first-order difference equation

$$\mathbf{x}_{k+1} = P\mathbf{x}_k \quad \text{for } k = 0, 1, 2, \dots$$

If P is a stochastic matrix, then a **steady-state vector** (or **equilibrium vector** or **invariant probability vector**) for P is a probability vector $\mathbf{q}$ such that

$$P\mathbf{q} = \mathbf{q}$$

In fact the steady-state vector is the eigenvector of P that is corresponding to the eigenvalue of 1

Consider: A probability vector $\vec{x}_0$ and a stochastic matrix P such that the Markov chain $\vec{x}_k = P^k \vec{x}_0$, $k \geq 0$, has no limit as k tends to infinity.

$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ s.s. vector: $\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$
if $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, then there is no limit.

probability vector.

Q: When will $\underline{x}_0$ always converges to the steady state vector for any choice of $\underline{x}_0$?

A: regular stochastic matrix P .

A stochastic matrix P is **regular** if some power P^k contains only strictly positive entries.

If P is a regular $m \times m$ transition matrix with $m \geq 2$, then the following statements are all true.

- There is a stochastic matrix Π such that $\lim_{n \rightarrow \infty} P^n = \Pi$.
- Each column of Π is the same probability vector $\mathbf{q}$.
- For any initial probability vector $\mathbf{x}_0$, $\lim_{n \rightarrow \infty} P^n \mathbf{x}_0 = \mathbf{q}$. $\leftarrow$ S.S. vector
- The vector $\mathbf{q}$ is the unique probability vector which is an eigenvector of P associated with the eigenvalue 1.
- All eigenvalues λ of P other than 1 have $|\lambda| < 1$.

When $\lim_{n \rightarrow \infty} x_n = q$, the meaning of q is:

- ① long-run probability
- ② occupation times

Google Matrix:

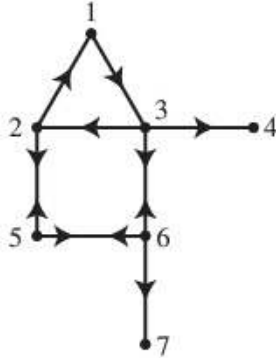


FIGURE 1

A seven-page Web.

Adjustment 1:

Handling the dangling node 4 & 7

$$P = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 0 & 1/2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 0 & 1/2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1/3 & 0 \\ 0 & 0 & 1/3 & 1 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 & 0 & 1/3 & 0 \\ 0 & 0 & 1/3 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/3 & 1 \end{bmatrix}$$

$$P_* = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 0 & 1/2 & 0 & 1/7 & 0 & 0 & 1/7 \\ 0 & 0 & 1/3 & 1/7 & 1/2 & 0 & 1/7 \\ 1 & 0 & 0 & 1/7 & 0 & 1/3 & 1/7 \\ 0 & 0 & 1/3 & 1/7 & 0 & 0 & 1/7 \\ 0 & 1/2 & 0 & 1/7 & 0 & 1/3 & 1/7 \\ 0 & 0 & 1/3 & 1/7 & 1/2 & 0 & 1/7 \\ 0 & 0 & 0 & 1/7 & 0 & 1/3 & 1/7 \end{bmatrix}$$

Adjustment 2: Damping factor $p=0.85$

$$G = p \cdot P_* + (1-p)E$$

$$E = \frac{1}{n} \cdot (n \times n \text{ matrix of } 1\text{'s})$$

Google
Matrix

$$G = .85 \begin{bmatrix} 0 & 1/2 & 0 & 1/7 & 0 & 0 & 1/7 \\ 0 & 0 & 1/3 & 1/7 & 1/2 & 0 & 1/7 \\ 1 & 0 & 0 & 1/7 & 0 & 1/3 & 1/7 \\ 0 & 0 & 1/3 & 1/7 & 0 & 0 & 1/7 \\ 0 & 1/2 & 0 & 1/7 & 0 & 1/3 & 1/7 \\ 0 & 0 & 1/3 & 1/7 & 1/2 & 0 & 1/7 \\ 0 & 0 & 0 & 1/7 & 0 & 1/3 & 1/7 \end{bmatrix}$$

$$+ .15 \begin{bmatrix} 1/7 & 1/7 & 1/7 & 1/7 & 1/7 & 1/7 & 1/7 \\ 1/7 & 1/7 & 1/7 & 1/7 & 1/7 & 1/7 & 1/7 \\ 1/7 & 1/7 & 1/7 & 1/7 & 1/7 & 1/7 & 1/7 \\ 1/7 & 1/7 & 1/7 & 1/7 & 1/7 & 1/7 & 1/7 \\ 1/7 & 1/7 & 1/7 & 1/7 & 1/7 & 1/7 & 1/7 \\ 1/7 & 1/7 & 1/7 & 1/7 & 1/7 & 1/7 & 1/7 \\ 1/7 & 1/7 & 1/7 & 1/7 & 1/7 & 1/7 & 1/7 \end{bmatrix}$$

$$= \begin{bmatrix} .021429 & .446429 & .021429 & .142857 & .021429 & .021429 & .142857 \\ .021429 & .021429 & .304762 & .142857 & .446429 & .021429 & .142857 \\ .871429 & .021429 & .021429 & .142857 & .021429 & .304762 & .142857 \\ .021429 & .021429 & .304762 & .142857 & .021429 & .021429 & .142857 \\ .021429 & .446429 & .021429 & .142857 & .021429 & .304762 & .142857 \\ .021429 & .021429 & .304762 & .142857 & .446429 & .021429 & .142857 \\ .021429 & .021429 & .021429 & .142857 & .021429 & .304762 & .142857 \end{bmatrix}$$

$\Rightarrow$

$$q = \begin{bmatrix} .116293 \\ .168567 \\ .191263 \\ .098844 \\ .164054 \\ .168567 \\ .092413 \end{bmatrix}$$

$\hookrightarrow$ regular!

Page rank:

$3 > 2 = 6 > 5 > 1 > 4 > 7$

Diagonalization of Symmetric Matrices

The Spectral Theorem for Symmetric Matrices

An $n \times n$ symmetric matrix A has the following properties:

- A has n real eigenvalues, counting multiplicities.
- The dimension of the eigenspace for each eigenvalue λ equals the multiplicity of λ as a root of the characteristic equation.
- The eigenspaces are mutually orthogonal, in the sense that eigenvectors corresponding to different eigenvalues are orthogonal.
- A is orthogonally diagonalizable.

Spectral Decomposition

$$A = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^T + \cdots + \lambda_n \mathbf{u}_n \mathbf{u}_n^T$$

$$A = \underset{P}{[\mathbf{u}_1 \cdots \mathbf{u}_n]} \underset{D}{\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}} \underset{P^T}{\begin{bmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_n \end{bmatrix}}$$

Quadratic Forms

$$A = \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix} \quad \text{matrix of quadratic form.}$$

Quadratic Form

$$\begin{aligned} \mathbf{x}^T A \mathbf{x} &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3x_1 - 2x_2 \\ -2x_1 + 7x_2 \end{bmatrix} \\ &= x_1(3x_1 - 2x_2) + x_2(-2x_1 + 7x_2) \\ &= 3x_1^2 - 2x_1x_2 - 2x_2x_1 + 7x_2^2 \\ &= 3x_1^2 - 4x_1x_2 + 7x_2^2 \end{aligned}$$

Change of variable in quadratic form:

The Principal Axes Theorem

Let A be an $n \times n$ symmetric matrix. Then there is an orthogonal change of variable, $\mathbf{x} = P\mathbf{y}$, that transforms the quadratic form $\mathbf{x}^T A \mathbf{x}$ into a quadratic form $\mathbf{y}^T D \mathbf{y}$ with no cross-product term.

$$\begin{aligned} \mathbf{y} &= P^{-1} \mathbf{x} = P^T \mathbf{x} \Rightarrow (P^T \mathbf{x})^T D P^T \mathbf{x} = \mathbf{x}^T P D P^T \mathbf{x} = \mathbf{x}^T A \mathbf{x} \\ &\Rightarrow A = P D P^T \end{aligned}$$

Categories of Quadratic Forms

A quadratic form Q is:

- a. **positive definite** if $Q(\mathbf{x}) > 0$ for all $\mathbf{x} \neq \mathbf{0}$,
- b. **negative definite** if $Q(\mathbf{x}) < 0$ for all $\mathbf{x} \neq \mathbf{0}$,
- c. **indefinite** if $Q(\mathbf{x})$ assumes both positive and negative values.

Quadratic Forms and Eigenvalues

Let A be an $n \times n$ symmetric matrix. Then a quadratic form $\mathbf{x}^T A \mathbf{x}$ is:

- a. positive definite if and only if the eigenvalues of A are all positive,
- b. negative definite if and only if the eigenvalues of A are all negative, or
- c. indefinite if and only if A has both positive and negative eigenvalues.

Constrained Optimization

Let A be a symmetric matrix, and define m and M as in (2). Then M is the greatest eigenvalue λ_1 of A and m is the least eigenvalue of A . The value of $\mathbf{x}^T A \mathbf{x}$ is M when $\mathbf{x}$ is a unit eigenvector $\mathbf{u}_1$ corresponding to M . The value of $\mathbf{x}^T A \mathbf{x}$ is m when $\mathbf{x}$ is a unit eigenvector corresponding to m .

$$m = \min \{ \mathbf{x}^T A \mathbf{x} : \|\mathbf{x}\| = 1 \}, \quad M = \max \{ \mathbf{x}^T A \mathbf{x} : \|\mathbf{x}\| = 1 \} \quad (2)$$

Let A , λ_1 , and $\mathbf{u}_1$ be as in Theorem 6. Then the maximum value of $\mathbf{x}^T A \mathbf{x}$ subject to the constraints

$$\mathbf{x}^T \mathbf{x} = 1, \quad \mathbf{x}^T \mathbf{u}_1 = 0$$

is the second greatest eigenvalue, λ_2 , and this maximum is attained when $\mathbf{x}$ is an eigenvector $\mathbf{u}_2$ corresponding to λ_2 .

Let A be a symmetric $n \times n$ matrix with an orthogonal diagonalization $A = PDP^{-1}$, where the entries on the diagonal of D are arranged so that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ and where the columns of P are corresponding unit eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_n$. Then for $k = 2, \dots, n$, the maximum value of $\mathbf{x}^T A \mathbf{x}$ subject to the constraints

$$\mathbf{x}^T \mathbf{x} = 1, \quad \mathbf{x}^T \mathbf{u}_1 = 0, \quad \dots, \quad \mathbf{x}^T \mathbf{u}_{k-1} = 0$$

is the eigenvalue λ_k , and this maximum is attained at $\mathbf{x} = \mathbf{u}_k$.

The Singular Value Decomposition

The Singular Value Decomposition

Let A be an $m \times n$ matrix with rank r . Then there exists an $m \times n$ matrix Σ as in (3) for which the diagonal entries in D are the first r singular values of A , $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$, and there exist an $m \times m$ orthogonal matrix U and an $n \times n$ orthogonal matrix V such that

$$A = \underbrace{U}_{\text{Orthonormal}} \underbrace{\Sigma}_{\text{diagonal}} \underbrace{V^T}_{\text{Orthonormal}}$$

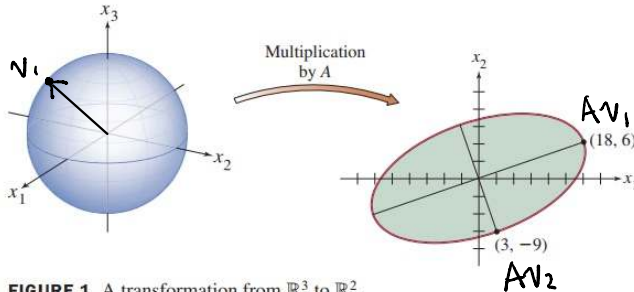
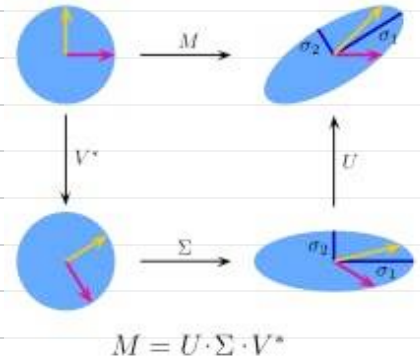
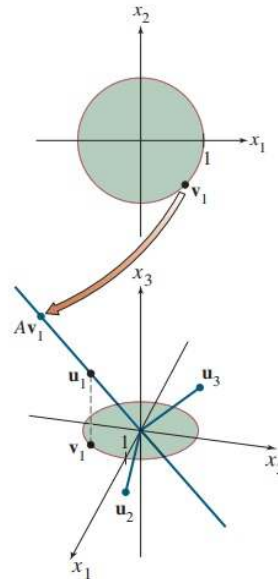


FIGURE 1 A transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

$$A = \begin{matrix} 2 \times 3 \\ U \end{matrix} \begin{matrix} 2 \times 2 \\ \Sigma \end{matrix} \begin{matrix} 2 \times 3 \\ V^T \end{matrix} \begin{matrix} 3 \times 3 \end{matrix}$$



Consider the
constrained optimization
problem:
 $\max/\min \|Ax\|$
when $\|x\|=1$

$$\begin{aligned} \|Ax\|^2 &= (Ax)^T(Ax) = x^T A^T A x \\ &= x^T (A^T A) x \end{aligned}$$

Recipe for finding SVD of A ($m \times n$, rank r)

① Find the orthogonal diagonalization of $A^T A$

$$A^T A = V D V^T$$

$n \times n$ $\uparrow$ $\uparrow$ $\uparrow$
 $n \times n$ $n \times n$ $n \times n$

② Find the matrices V & Σ

order the eigenvalues in D in the decreasing order $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r \geq \dots \geq \lambda_n$
Correspondingly order the eigenvalues and normalize them to form $V \leftarrow n \times n$
compute singular values: $\sigma_i = \sqrt{\lambda_i}$ to form the diagonals of $\Sigma \leftarrow m \times n$

$$\lambda_{r+1} = \dots = \lambda_n = 0$$

$$\Sigma = \begin{bmatrix} \lambda_1 & & 0 & & \\ & \ddots & & & \\ 0 & & \lambda_r & & 0 \\ & & & \ddots & \\ 0 & & 0 & & 0 \end{bmatrix} = \begin{bmatrix} r \times r & r \times (n-r) \\ (m-r) \times r & (m-r) \times (n-r) \end{bmatrix}$$

$$V = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_n \\ | & | & & | \end{bmatrix}$$

③ Construct the matrix U

$$\text{for } i=1, \dots, r, \quad u_i = \frac{1}{\sigma_i} A v_i$$

for $i=r+1, \dots, m$, $\{u_{r+1}, \dots, u_m\}$ is an orthonormal basis of $(\text{Span}\{u_1, \dots, u_r\})^\perp$

Find a basis of $(\text{Span}\{u_1, \dots, u_r\})^\perp$ then use Gram-Schmidt to find the orthogonal basis and finally normalize them

$$U = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \dots & u_m \\ | & | & & | \end{bmatrix} \leftarrow m \times m.$$

Conditional Number

If A is an invertible $n \times n$ matrix, then the ratio σ_1/σ_n of the largest and smallest singular values gives the **condition number** of A . Exercises 50–52 in Section 2.3 showed how the condition number affects the sensitivity of a solution of $Ax = b$ to changes (or errors) in the entries of A . (Actually, a “condition number” of A can be computed in several ways, but the definition given here is widely used for studying $Ax = b$.) ■